

Imaging-based single-cell spatial data: Xenium

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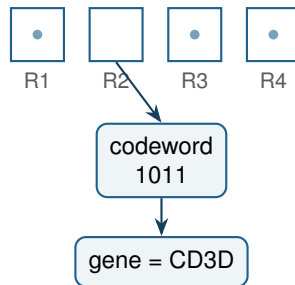
Yesterday: spot-based Visium, the standard workflow, and first spatial statistics. A spot mixes several cells.

Today we move to **imaging-based** data at single-cell resolution. You already know spatial objects and the scanpy/squidpy workflow, so we focus on what is *new* with Xenium:

- the data: individual molecules and a per-cell segmentation, not a count matrix alone;
- segmentation as the step that everything downstream inherits;
- QC and embeddings specific to sparse, single-cell panels;
- the same spatial statistics, now between single cells.

How imaging-based ST works

- A targeted gene panel is detected *in situ* by cyclic hybridisation and imaging of fluorescent probes.
- Each gene has a code; across rounds the code is read out and decoded to a transcript at an (x, y) location.
- Transcripts are assigned to cells using a segmentation mask.
- Output: transcripts, a cell $\times$ gene matrix, cell metadata, and images.



Imaging-based platforms

Platform	Panel size	Note
MERFISH / Vizgen	~300–1000	error-robust barcodes
seqFISH+	~10,000	many pseudocolours
Xenium (10x)	280–480, up to 5000 (Prime)	in situ, H&E compatible
CosMx (NanoString)	~1000–6000	RNA and protein

Panel size is a design decision. A focused 480-gene panel (we use the 377-gene Multi-Tissue and Cancer panel) targets known markers at high sensitivity; the 5,000-plex *Xenium Prime* trades some per-gene sensitivity for breadth. Either way you only see genes on the panel, so the panel bounds the biology.

Xenium outputs: the files

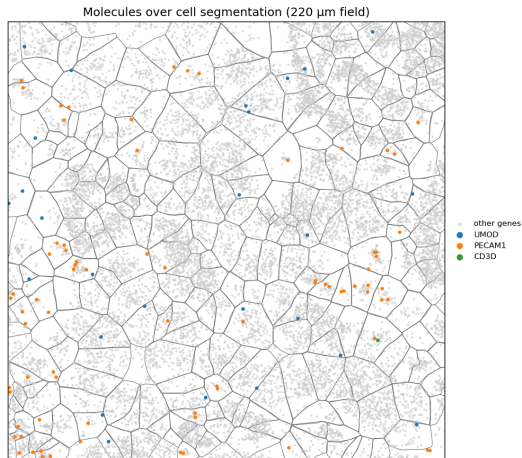
- `cell_feature_matrix.h5`: cell $\times$ feature counts (genes *and* controls).
- `cells.parquet`: centroid, area, counts per cell.
- `transcripts.parquet`: every decoded molecule with (x, y, z) , gene, and a quality value `qv`.
- `*_boundaries.parquet`: cell and nucleus polygons.
- `morphology_*.ome.tif`: images (large).

Our kidney section: **97,560 cells**, **377 panel genes**, **~ 8.1 M** decoded transcripts.

Reading path.

`spatialdata_io.xenium` builds a full object with images and shapes (great to visualise, heavy). For the table analysis we read the matrix and cell table straight into `AnnData`: same cells, a fraction of the memory. We pull in the molecules and boundaries only for a small field.

The molecule layer: what makes Xenium different



Unlike a Visium spot, Xenium localises *individual transcripts* and assigns each to a cell through a segmentation polygon.

- Grey points are molecules; coloured points are marker genes; outlines are the per-cell boundaries.
- You can see expression *build up* the count matrix, and where a transcript sits relative to a boundary.
- Errors enter here: a molecule near an edge can be assigned to the wrong cell.

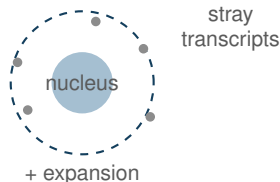
Crop from the kidney section; transcripts at $qv \geq 20$.

Segmentation is the crux

Cells are defined by a mask, and everything downstream inherits its errors.

- In dense tissue, boundaries are ambiguous: transcripts are misassigned, and small or nucleus-less cells are missed.
- Symptoms: implausible co-expression, “cells” with too few transcripts, lineage markers bleeding across boundaries.

Treat cell-level results as estimates and sanity-check against the raw transcript image where it matters.



Segmentation methods

- **Default (10x):** nuclear stain plus $\sim 15\ \mu\text{m}$ expansion; a multimodal deep-learning kit uses membrane/boundary stains when available.
- **Image-based:** Cellpose (generalist nucleus/membrane segmentation).
- **Transcript-based:** Baysor and Proseg learn transcript colocalisation; Proseg uses a cellular Potts model.
- **Learned, reference-aware:** segger (graph neural network, transcript to cell link prediction, can use scRNA-seq).

The default expansion explains the tessellated outlines on the previous slide. Benchmarks show a tradeoff between transcript capture and cell purity that is tissue-dependent; there is no universal winner.

Stringer et al., *Nat Methods* 2021 (Cellpose); Petukhov et al. (Baysor); Jones et al. 2025 (Proseg); Heidari et al. 2025 (segger).

Why segmentation matters for your biology

- Misassigned transcripts directly corrupt cell typing and, especially, cell–cell communication (a ligand can “jump” to the wrong cell).
- Nucleus-anchored methods miss cells without a captured nucleus, biasing composition.
- The default $\sim 15\text{ }\mu\text{m}$ expansion is a fixed guess at cell size, so it over- or under-shoots in tissues with very small or very large cells.
- Improving identity in poorly segmented, dense regions is an active and unsolved problem.

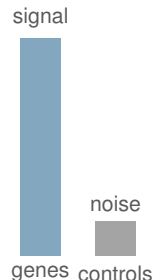
Heidari et al. 2025 (segger); CRISP segmentation benchmark, bioRxiv 2026.

Control features

The panel ships with built-in negative controls:

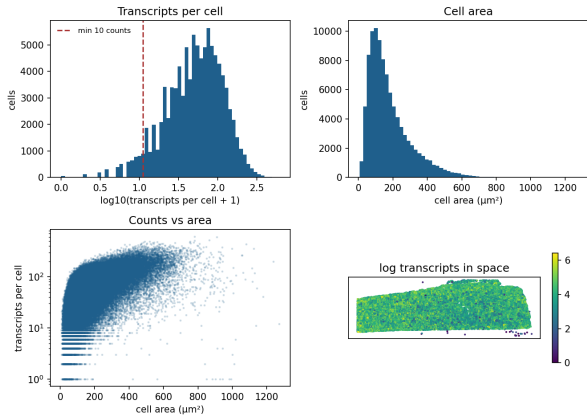
- **Negative control probes:** target no real transcript; estimate non-specific binding.
- **Negative control codewords (blanks):** valid codes assigned to no gene; estimate the decoding error rate.

A healthy run puts only a small fraction of counts on controls. A high control fraction in a cell or region is a warning.



Quality control and what to look for

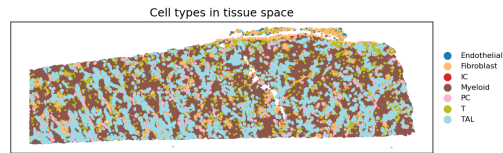
- **Transcripts / cell, genes / cell:** too few means an empty or fragmented segment.
- **Control fraction:** flags noise.
- **Cell area:** very small or large segments are suspect.
- **Counts vs area:** density should be roughly consistent.
- **Sensitivity:** counts are far below scRNA-seq, so a gene's absence is weak evidence.



Read every metric *in space*, not only as a histogram.

Cell typing on a small panel

- Fewer genes make clusters noisier and rare types harder to separate.
- Use a curated marker set for the tissue (here: kidney nephron segments, stroma, immune), then refine by eye and check spatial plausibility.
- Reference label transfer (CellTypist, scANVI) scales with a matched reference, but inherits segmentation contamination.
- BANKSY can type cells and find domains together using the neighbourhood.



Marker-scored cell types laid back onto the tissue:
tubules, stroma, and immune cells recover anatomy.

Domínguez Conde et al., *Science* 2022 (CellTypist);
Singhal et al., *Nat Genet* 2024 (BANKSY).

The UMAP is a *summary*, not a measurement, and many choices move it. They fall into two groups, and we will see which ones actually matter.

Data factors

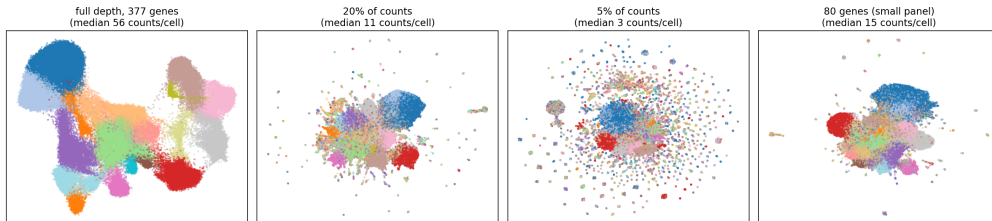
- depth (counts per cell)
- panel size (number of genes)
- QC / filtering thresholds
- normalisation

Method factors

- number of principal components
- neighbours k , `min_dist`, distance metric
- clustering resolution, random seed

Data factors: depth and panel size

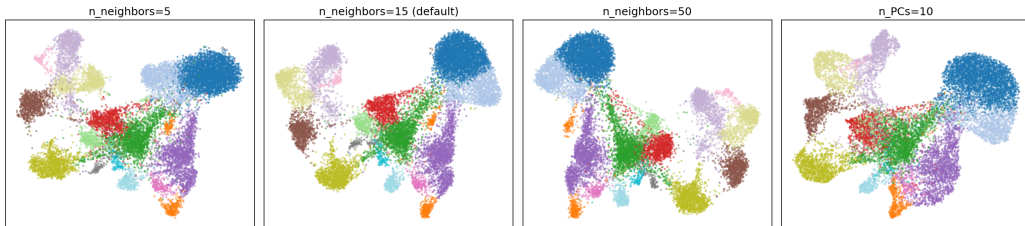
Same cells, coloured by full-depth clusters: separation degrades with depth and panel size



Same cells, coloured by the full-depth clusters. Thinning the counts (binomial downsampling) or shrinking the panel collapses well-separated clusters toward one blob, the regime where UMAPs also show thin streaks. The biology has not changed, the available information has.

Method parameters reshape the same data

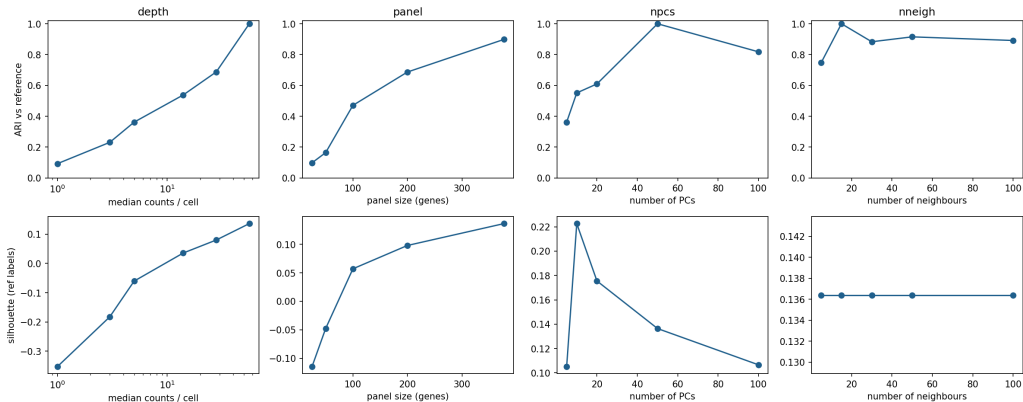
Same cells and counts, coloured by the reference clusters: method parameters reshape the embedding



Identical cells and counts; only the embedding settings differ. Small k (`n_neighbors`) exaggerates local structure and fragments clusters; large k smooths everything toward one blob; too few PCs discard signal. The picture changes a lot while the biology does not, so never read cluster sizes, gaps, or between-cluster distances off a UMAP as if they were quantitative.

Systematic analysis: quantifying the impact

What shapes the embedding: partition stability (ARI) and class separation (silhouette) vs each factor



One factor varied at a time on the real section. **ARI** (top): partition stability vs the full-data reference. **Silhouette** (bottom): separation of the reference cell types in PC space.

Reading the systematic analysis

- **Depth and panel size dominate.** As counts per cell or gene count fall, both metrics drop steeply and the silhouette turns negative: the reference cell types stop being separable.
- **Number of PCs is second-order and non-monotonic.** Too few PCs underfit; too many re-inject noise. Silhouette is best around 10–20 PCs.
- **Neighbours k reshapes the graph, not the PC space.** It changes the clustering (ARI) but leaves the silhouette flat, so it cannot rescue separation the data do not contain.
- **Takeaway.** Spend effort upstream (counts, panel, QC, normalisation); method knobs tune the picture but cannot add information that is not there.

Quantifying it: definitions and the noise floor

Adjusted Rand Index compares two clusterings (conditioned vs reference), corrected for chance:

$$\text{ARI} = \frac{\sum_{ij} \binom{n_{ij}}{2} - [\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}] / \binom{n}{2}}{\frac{1}{2} [\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}] - [\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}] / \binom{n}{2}},$$

1 for identical partitions, ≈ 0 for chance.

Silhouette of cell i , with mean within-class distance a_i and nearest other-class mean b_i :

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)} \in [-1, 1],$$

averaged over cells; higher means better-separated types.

Why depth dominates. Counts are roughly Poisson: for a gene with mean λ , $\text{CV} = \sigma/\mu = 1/\sqrt{\lambda}$. Halving depth raises relative noise by $\sqrt{2}$, and $\log_{10} p$ of small integers is discrete, so distances between sparse cells reflect sampling noise more than biology.

Choosing a spatial graph



kNN



Delaunay



radius

kNN fixes the neighbour count; Delaunay follows tissue geometry; radius fixes a physical distance. The choice changes enrichment results, so state it.

- **Neighbourhood enrichment:** which cell types co-locate vs avoid each other (permutation test on the graph).
- **Co-occurrence:** how that association changes with distance, separating tight contact from regional association.
- **Ripley's K / L:** clustering vs dispersion of a cell type across scales.
- **Spatially variable genes:** Moran's I over cells.

These are the building blocks for the organisation analysis on Day 3.

Interactive exploration: napari and Xenium Explorer

Static plots only go so far for molecule-level data. To explore freely:

- **napari** with **napari-spatialdata**: layer the morphology image, cell/nucleus polygons, transcripts, and any `obs` column; pan, zoom, and toggle layers.
- **10x Xenium Explorer**: the vendor viewer for the raw bundle.
- Use it to audit segmentation in a dense region and to check a marker sits where you expect before trusting a cell-level result.

This is a *local*, interactive step: it needs a desktop session and is not run on the headless lab server.

The notebook includes an optional, guarded cell that opens the section in napari when you run it on your own machine.

Marconato et al., *Nat Methods* 2025
(SpatialData / napari-spatialdata).

SpatialData: when it is worth it

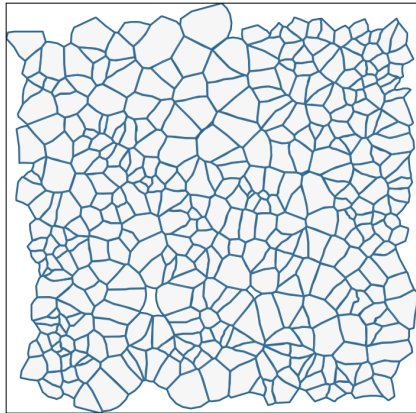
SpatialData aligns images, shapes (boundaries), points (transcripts), and tables in one lazy object with a shared coordinate system.

Worth it for image and molecules together: visualising or re-doing segmentation, cropping aligned multimodal regions, multi-sample registration.

Overkill for table-level work (QC, clustering, typing, neighbourhood statistics), where plain `AnnData` is lighter, so today's workflow stays on `AnnData`.

Marconato et al., *Nat Methods* 2025 (SpatialData).

SpatialData: cell boundaries (rendered with `spatialdata-plot`)



Boundaries via `spatialdata-plot` (notebook 02b).

Beyond one section

- Multiple sections bring batch effects. Integration with Harmony or scVI aligns embeddings before joint clustering.
- Spatial-aware methods (e.g. BANKSY) can do batch-aware domain detection.
- Comparing conditions needs composition-aware tests, not just eyeballing proportions (Day 3).

Korsunsky et al., *Nat Methods* 2019 (Harmony); Lopez et al., *Nat Methods* 2018 (scVI).

Open problems on imaging-based data

- Segmentation in dense tissue, and its propagation into typing and communication.
- Low detection of cytokines and chemokines, so immune states are hard to identify.
- Reliability of spatial QC metrics, which are still being standardised.
- Integrating images (H&E) with the molecular layer for better region identity.

Further reading

- Petukhov et al. Baysor: cell segmentation from transcripts.
- Jones et al. 2025 (Proseg); Heidari et al. 2025 (segger); Stringer et al., *Nat Methods* 2021 (Cellpose).
- Singhal et al. BANKSY. *Nat Genet* 2024.
- Marconato et al. SpatialData. *Nat Methods* 2025.
- 10x Genomics analysis guides for Xenium segmentation and QC; Squidpy Xenium tutorials: squidpy.readthedocs.io.

Dataset: a public Xenium non-diseased human kidney section ($\sim 97,560$ cells).

You will:

- read the section the lean way (matrix + cell table);
- look at molecules over cell boundaries on a crop (and, optionally, SpatialData / napari);
- separate control features and compute Xenium QC;
- filter, normalise, cluster, and annotate cell types with markers;
- build a spatial graph and run enrichment, co-occurrence, and Moran's I.

Exercises: set thresholds and map removed cells, refine annotation, compare kNN vs Delaunay graphs, and interpret spatially variable genes.